SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE:

CSA5116

COURSE OUTCOME COVERED

C01: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:**50**

Annexure A

CONCEPT MAPPING

Code of *Conduct.*

*I Sarika.K S(*192521039**) (***Name/Reg No*) *certify that this submission* is *my original* work *and that I* have *adhered to* the *guidelines specified for this assessment*.

I understand that any violation of *academic integrity rules* will *result in disciplinary action*.

Sarika. KS

Signature* of the *student:

CONCEPT MAPPING

Security Attacks. Threats, Risks and security Goals

Security Attacks

(eg. phirling, Data modification, DDOS)

sewvity Goals

(confidentiality, integsity, Avaliability)

1. confidentiality

Howeats

Leg. Information <hoft,

Unauthorized alteration sowie discption]

Pirks

Leg. Data disclosure

system unavailable

[phishing) unauthorized acr

↓

Attacker aktains confidential Information

parri aids or porronal data may be stalen confidentiality

Encryption, authentication, acers contral

Real world Example:

A phishing email steals a weis banking parew on and `expores` pouvate account Information

2. Intergsity

Data modification / main-`in`-the middle attack)

unauthorized perron changes data

`Incorrect` or manipulated Information

Integrity

Harling, digital signatures, merrage authentication

Real would example:

An attacked changes an online payment amount form 11000 to 10,000

Avaliability

1,000

Denial- of service (DOS) DDOS)

attack

System verowie become unavailable

urers cannot acers the service

هیچ

Availability

Firewalls, to affic, filtering,
backup systems,

Real would Example:

A banking webrite becomes unavailable because

attackers wend a huge amount of traffic

Commbrion:-

Security attacks create threats, which produce risks to Information systems. The there majors security goals - confidentiality, Integrity and availability (CIA) -help vedure there würks thorough appropriate security.

mechanisms.

4) Concept map -

Steganography

steganography

Data Hiding

watermarking

suwivability

1. Data Hiding

Meaning:

servie communication

conceals secret Information inside another orinary looking medium Quch as an

Relationship:-

notice.

an Image, audio or video

seviet Data > Hidden in cover media > Difficult to

contribution:-

Hides the existence of the communication, making

it difficult for an attacker to detect the secret merrage.

Real would application:

A confidential merrage can be hidden inside an image hand bent through a normal communication channel

2. Watermarking:-

meaning:.

Embeds information into digital media to Identify ownership, authenticity or copyright

Relationship : –

Digital content -> watermark -> ownworhip | Authentication ->

3. Survivability:-

Meaning:

a watermark to remain

Ability of hidden Information or available even after the media is modified comprereed.

Relationship:

Hidden Information - modification/ compression s Information so wowvives.

Kontribution:-

Makes hidden information more roburt against operations like compressions, verizing or format

conversion. Real-world application:-

A watermaak in a digital Image can remain detectable even after the image is compressed and sharied online

Condurion.-

Data hidden facures on concealing information, (watermarking facures on ownership and authenticity land survivability ensures that hidden information remains detectable after modification Together, the improve Secure and ocliable digital Communication

RUBRICS FOR CALCULATION:

Criteria	Weighta **Level 4** – ge **Exemplary**	Level 3- Proficient	Basic
	Demonstrates thorough	Good understandi	

				Level 2- Developing	Developing Level **1**
				understanding;	understanding Beginning Poor
Understandin g of Concepts	25	understanding and effectively integrates multiple ogies	**ng** with proper	**limited** or partial	; unable to integrate
			concepts Correct		
Experiment al Design	30	concepts/technol Well-planned, systematic implementation with optimal solution	integration of most		

Basic

Incorrect **or**

implementati implementatio incomplete

on with minor issues

n with errors or

inefficiencies

Limited or

inconsistent use of tools

implementatio

n

Incorrect **or**

minimal use

of tools

integration

concepts

Effective and

Appropriate

advanced use of

use of tools

Use of

Tools

20

tools/technologies with correct

with proper

execution

configuration

Insightful analysis

Good

Basic

with accurate

analysis

analysis with

incorrect

Analysis of Results

interpretation and

with

limited

15

validation of

reasonable insights

Poor **or**

analysis of results

results

interpretatio n

Clear, well-

Organized

Basic

Poor **or**

structured

documentatio documentatio

incomplete

Documentati on

10

documentation

n with

n **with**

documentatio

with diagrams, code, and results

necessary details

missing details

n

L